

Learning AI By Hand : Information Retrieval (AI BY HAND inspired)

Daril Vergesse Sobze Soffack

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1. Exercise

Query : *"What are the AI Engineer skills?"*

Tokens : *[ai, engineer, skills]*

Documents :

- **Doc 1 (construction architect) :** *Construction architects need skills in structural design, building regulations. They have also a talent in project management[...].*
- **Doc 2 (AI engineer) :** *AI engineers require strong skills in AI/ML, especially in AI model design, and they have to have software engineering talent in Python[...].*
- **Doc 3 (software engineer salaries) :** *The average salary for software engineers has increased, with strong demand in the tech industry[...].*

Goal : Calculate the cosine similarity between the query and each document using TF-IDF and identify the most relevant document.

2. Definitions

- **Term Frequency (TF) :**

$tf(t, d)$ = the number of times term t occurs in the document d

- **Normalized TF :**

$\text{normTF}(t, d) = \log(tf(t, d) + 1), \quad (0 \text{ si } tf(t, d) = 0)$

- **Document Frequency (DF) :**

$df(t)$ = the number of document in which t occurs

- Inverse Document Frequency (IDF) :

$$idf(t) = \log \frac{N}{df(t)}, \quad N = \text{total numbers of documents}$$

- TF-IDF :

$$tfidf(t, d) = \text{normTF}(t, d) \times idf(t)$$

NB: we won't use normTF in this exercise but rather the simple TF.

- Cosine Similarity :

$$\text{cosine_sim}(Q, D) = \frac{Q \cdot D}{\|Q\| \cdot \|D\|}$$

3. Solution

step 1 : usefull token per document

Document	presents Tokens
Doc1	skills
Doc2	ai, engineer, skills
Doc3	engineer

Step 2 : Term Frequency TF (thought up by myself for the 3 documents)

Token	TF				DF	IDF	TF-IDF			
	q	d1	d2	d3	df	idf	q	d1	d2	d3
AI	1	0	15	1	100,000	1	1	0	15	1
Engineer	1	0	10	15	10,000	2	2	0	20	30
Skills	1	10	10	0	100	4	4	40	40	0

Table 1: Term Frequencies in query and documents.

Step 3 : Document frequency and Inverse (thought up by myself)

$$df(AI) = 100000, \quad df(engineer) = 10000, \quad df(skills) = 100$$

$$idf(t) = \log \frac{N}{df(t)}, \quad N = \text{total numbers of documents} = 1000000$$

Token	TF				DF	IDF	TF-IDF			
	q	d1	d2	d3	df	idf	q	d1	d2	d3
AI	1	0	15	1	100,000	1	1	0	15	1
Engineer	1	0	10	15	10,000	2	2	0	20	30
Skills	1	10	10	0	100	4	4	40	40	0

Table 2: Document frequency and Inverse.

Step 4 : TF-IDF

Query and documents vectors:

$$tfidf(t, d) = \text{normTF}(t, d) \times idf(t)$$

NB: we won't use normTF in this exercise but rather the simple TF.

$$Q = [AI = 1, engineer = 2, skills = 4]$$

$$Doc1 = [AI = 0, engineer = 0, skills = 40]$$

$$Doc2 = [AI = 15, engineer = 20, skills = 40]$$

$$Doc3 = [AI = 1, engineer = 30, skills = 0]$$

Token	TF				DF	IDF	TF-IDF			
	q	d1	d2	d3	df	idf	q	d1	d2	d3
AI	1	0	15	1	100,000	1	1	0	15	1
Engineer	1	0	10	15	10,000	2	2	0	20	30
Skills	1	10	10	0	100	4	4	40	40	0

Table 3: TF-IDF for query and documents.

Step 5 : Cosine Similarity

$$\|Q\| = \sqrt{1^2 + 2^2 + 4^2} \approx 4.58$$

$$\|Doc1\| = \sqrt{0^2 + 0^2 + 40^2} = 40$$

$$\|Doc2\| = \sqrt{15^2 + 20^2 + 40^2} \approx 47, 17$$

$$\|Doc3\| = \sqrt{1^2 + 30^2 + 0^2} = 30$$

Dot products :

$$Q \cdot Doc1 = 160, \quad Q \cdot Doc2 = 215, \quad Q \cdot Doc3 = 60$$

Cosine similarities :

$$\text{cosine_sim}(Q, D) = \frac{Q \cdot D}{\|Q\| \cdot \|D\|}$$

$$\text{Doc1} \approx 0.87, \quad \text{Doc2} \approx 0.995, \quad \text{Doc3} \approx 0.436$$

4.final Results

Document	Cosine similarity	relevance
Doc2	0.995	most relevant
Doc1	0.87	somewhat relevant
Doc3	0.436	irrelevant

Conclusion :

Doc2 is the most relevant for the query “AI Engineer skills.”